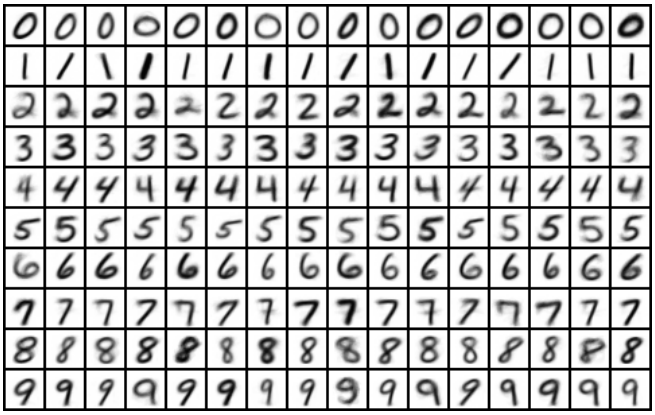

Figure 7. For every digit, we perform K-means clustering with K=16. We show here the cluster centroids, which intuitively are anchor images within the manifold of all possible images.

from the last fully connected LeNet layer for the synthesized examples, as well as for the held out test examples, and compute FID score as normal, by computing the Wasserstein distance between the Gaussians. Note that we use embeddings for LeNet trained on MNIST, rather than the Inception network trained on ImageNet, because MNIST isn't comprised of natural images. This is consistent with the convention in (Alemohammad et al., 2024).

For the self-correction operation, we compute the K-means clusters, with $K = 16$, once at the start of training. And we "correct" a synthesized motion by mapping it to the nearest cluster mean corresponding to its digit. In Figure 7 we present the clusters, and we present graphs of our FID scores across augmentation percentages in Figure 8.

E. Additional Human Motion Generation Qualitative Results

In Figures 9, 10, and 11, we present additional qualitative observations and analysis of our synthesized motions. We present more evidence that iterative fine-tuning with self-correction yields physically plausible motions comparable to the baseline, whereas iterative fine-tuning without self-correction yields motions that are incorrect for various reasons. See the captions of the referenced figures for analysis of some characteristic failure modes of the iterative fine-tuning loop *without* self-correction.

A technical note: for all figures, we render the motions from the same environment and camera position. We consolidate each render into the same image *without* resizing it. This means that if a figure appears larger relative to the others, the human moved closer to the camera. Some motions will have transparent frames of past positions; the more transparent the image, the farther back in the past it was in the motion sequence. Finally, in each figure, the text prompt for all generated motions was the same –the prompt being the one associated with the ground truth motion in the HumanML3D (Guo et al., 2022) training data, which we also visualize. Note that the coloring in the humanoid figures corresponds to the coloring in the graphs.

F. Additional Human Motion Generation Quantitative Results

See Figures 12, 13, 14 for results when the dataset size is $n \in \{64, 128, 256\}$ and the synthetic augmentation percentage is $\lambda \in \{0.25, 0.50, 0.75, 1.00\}$. And see Figures 15 and 16 for additional results on our iterative fine-tuning experiments when the dataset size is $n = 2794$ and the synthetic augmentation percentage is $\lambda \in \{0.05, 0.10, 0.15, 0.20, 0.25\}$. The graphs provide evidence across 17 experiment settings that our iterative fine-tuning procedure with self-correction yields better training performance than iterative fine-tuning with no self-correction for the motion synthesis task, in accordance with Theorem 4.3.

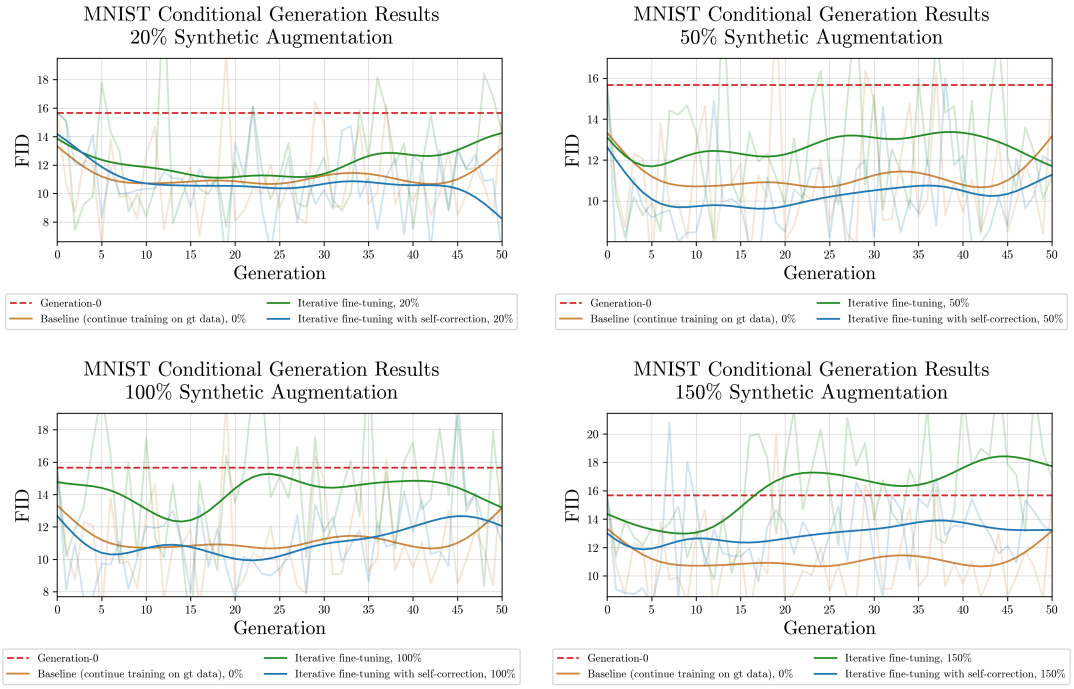

Figure 8. Results from MNIST experiments with iterative fine-tuning with and without self-correction. These graphs show the FID score on the last checkpoint for every generation; this is the checkpoint used for sampling in the self-consuming loop experiments, and it is also the checkpoint where training is resumed with this new partially synthesized dataset. These results demonstrate that **iterative fine-tuning with self-correction** generally outperforms **iterative fine-tuning.**

G. Consistency Across Seeds: Additional Human Motion Generation Quantitative Results

In Figures 17, 18, 19, and 20, we present experimental results from runs across three more seeds for our human motion experiments when the dataset size is $n = 64$. We find that the self-correction technique consistently yields improved training dynamics over iterative fine-tuning without correction.

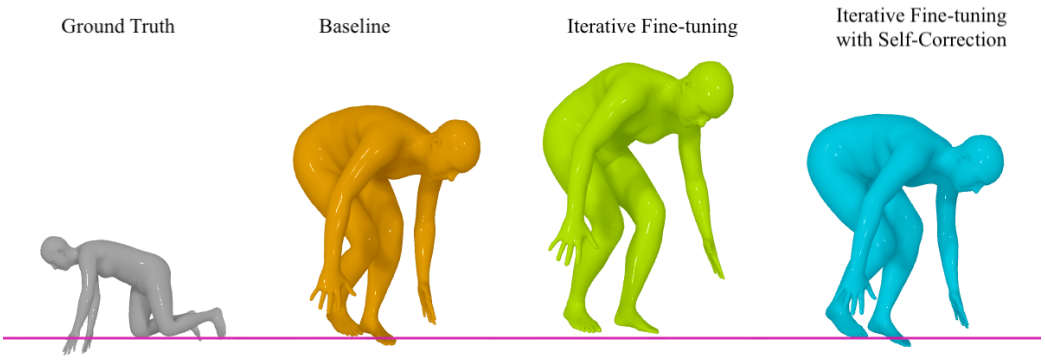

Figure 9. Here we see the negative *floating* phenomenon exacerbated by **iterative fine-tuning**, whereas **iterative fine-tuning with self-correction** generates a motion with floor contact integrity comparable to the **ground truth** and **baseline**. The floatic metric is formally defined in (Yuan et al., 2023) as the distance between the lowest vertex on the human mesh and the floor plane. All three sequences were generated using the same prompt: *person got down and is crawling across the floor.* Each snapshot was taken at exactly frame 87. The green figure appears larger than the other two only because it is closer to the camera. The two motions on the right were synthesized after 50 generations training with 25% synthetic augmentation, trained on $n = 64$ data points.

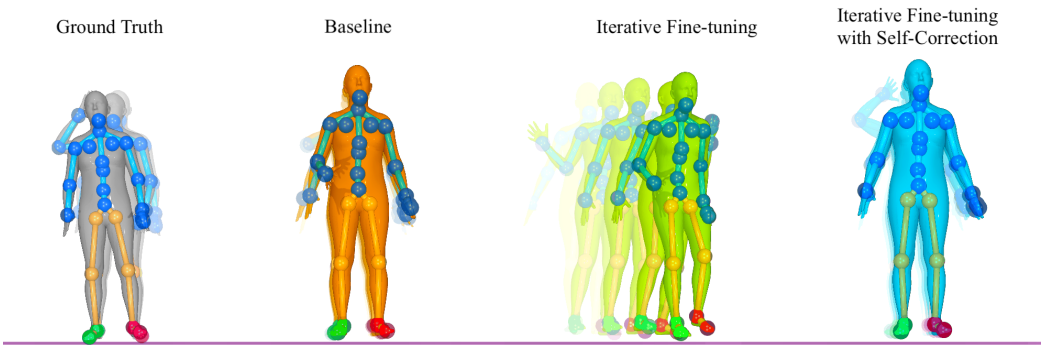

Figure 10. All four of the above motions correspond to the prompt: *a person raises right hand to face looks around and puts hand down back to side.*. The model which is trained with **iterative fine-tuning** outputs spurious motion that slides the figure to the right. And in the video for this example, the human rotates their forearm unnaturally and forcefully. In contrast, the **baseline** and **iterative fine-tuning with self-correction** models' motions both accurately embody the prompt. Each generated snapshot is taken at exactly frame 142 while the ground truth's image is frame 70 in its sequence. The two motions on the right were synthesized after 42 generations with 10% synthetic augmentation, where the ground truth dataset has size $n = 2794$.